

Credit Risk: Merton Structural Model

Folder: credit_risk | Survival probability, hazard rate, credit spread and CVA

1. Objective

Use Merton's structural model to turn a firm's balance-sheet and asset volatility into market-style credit quantities across maturities. The single notebook `credit_risk.ipynb` is closed-form (no PDE solver).

2. Model and inputs

The firm's asset value V follows geometric Brownian motion and default occurs if V is below the face value of debt D at maturity. Parameters used:

Parameter	Value
Assets today V_0	100
Debt D	50
Asset volatility σ	25%
Risk-free rate r	3%
Maturities	0.25, 0.5, 1, 3, 5, 7, 10, 15, 20, 35, 50 years

3. Quantities computed

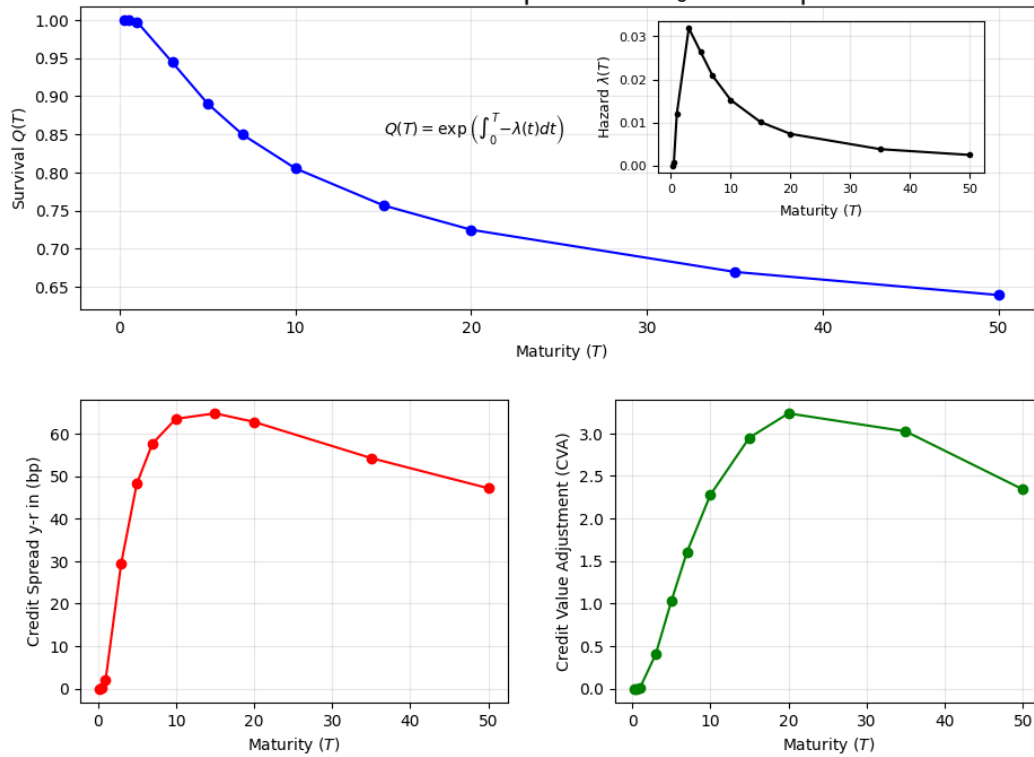
$$d_2 = [\ln(V_0/D) + (r - \sigma^2/2) T] / (\sigma \sqrt{T}), \quad d_1 = d_2 + \sigma \sqrt{T}$$

- **Survival probability** $Q(T) = \Phi(d_2)$ - the risk-neutral probability that $V_T > D$.
- **Hazard rate** $\lambda(T) = -d \ln Q / dT$, evaluated by central finite difference.
- **Credit spread** $y - r$ from $D e^{-yT} = e^{-rT} [D \Phi(d_2) + V_0 e^{rT} \Phi(-d_1)]$; a higher spread signals riskier debt.
- **CVA** $= D e^{-rT} \Phi(-d_2) - V_0 \Phi(-d_1)$, the amount a holder of the risky bond has overpaid relative to risk-free debt (the put-like value of the default option).

4. Output

A summary figure shows survival probability against maturity with the hazard rate as an inset, plus credit spread (in basis points) and CVA against maturity.

Merton Credit Risk Model | Assets $V_0 = 100$ | Debt $D = 50$



Merton model summary (credit_risk.png).

5. Files

File	Description
credit_risk.ipynb	All formulae, calculations and plots
credit_risk.png	Exported summary figure